Network Design with EIGRP, VLANs, DHCP, DNS and Routing Verification

Question: Design a network with 2 routers running EIGRP. Subnet the network appropriately. Configure VLANs and trunking. Setup DHCP with option configuration. Add DNS server with multiple records. Verify routing and test DNS resolution using nslookup simulation

OBJECTIVES

- To design and configure a network using EIGRP dynamic routing.
- To perform VLSM subnetting and assign IP addresses properly.
- To configure VLANs and trunking on switches for network segmentation.
- To set up DHCP with options for automatic IP allocation.
- To configure a DNS server with multiple A records for name resolution.
- To verify EIGRP routing and test DNS resolution using nslookup simulation.

Software Requirements

- 2× Cisco 1941 Router (Router0 and Router1)
- 2× Cisco 2960 Switch (Switch0 and Switch1)
- 4× PC endpoints (PC0, PC1, PC2, PC3)
- 1× Server (Server0 — DNS + DHCP)

THEORY

EIGRP (Enhanced Interior Gateway Routing Protocol)

EIGRP is an advanced Cisco distance-vector routing protocol that uses a composite metric based on bandwidth and delay to determine the best path. It employs the Diffusing Update Algorithm (DUAL) for fast, loop-free convergence. Unlike RIP, EIGRP sends only incremental updates when a topology change occurs, reducing unnecessary bandwidth usage. It supports VLSM, classless routing, and unequal-cost load balancing within a single Autonomous System (AS).

VLAN (Virtual Local Area Network)

A VLAN is a logical grouping of network devices regardless of their physical location. VLANs segment a network into multiple broadcast domains, improving security, reducing unnecessary traffic, and simplifying network management. Trunk ports using IEEE 802.1Q encapsulation carry traffic from multiple VLANs between switches and routers. The router-on-a-stick method uses sub-interfaces on a single router port to provide inter-VLAN routing.

DHCP (Dynamic Host Configuration Protocol)

DHCP automates IP address assignment to clients within a network. A DHCP server maintains a pool of available addresses and leases them to requesting devices. Along with IP addresses, DHCP can provide additional option configurations such as the default gateway, DNS server address, and domain name. Excluded address ranges prevent critical devices from being assigned dynamically.

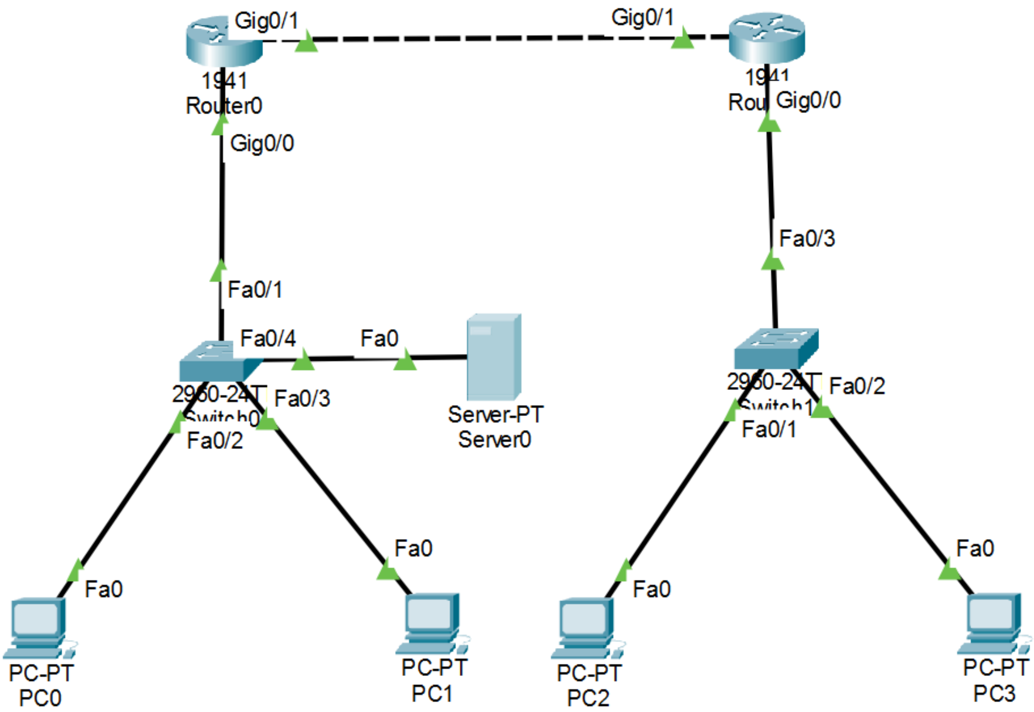
DNS (Domain Name System)

DNS is a hierarchical naming system that translates human-readable domain names into IP addresses. A DNS server maintains resource records such as A records (hostname to IP), which allow clients to resolve names without knowing numeric addresses. In Packet Tracer, a DNS server can be configured with multiple A records and verified using nslookup or ping commands that perform name resolution.

Trunking

A trunk link is a point-to-point connection between switches or between a switch and a router that carries traffic for multiple VLANs simultaneously. IEEE 802.1Q is the standard trunking protocol that adds a 4-byte VLAN tag to each Ethernet frame, allowing the receiving device to identify which VLAN the frame belongs to. Trunk ports must be configured to allow the desired VLANs explicitly.

NETWORK TOPOLOGY



CONFIGURATION

IP Addressing and VLAN Scheme

Device	Interface/ VLAN	IP Address	Subnet Mask	Default Gateway
Router0	Gig0/1 (WAN)	10.0.12.1	255.255.255.252	N/A
Router0	Gig0/0.10 (VLAN 10)	10.0.10.1	255.255.255.0	N/A
Router0	Gig0/0.20 (VLAN 20)	10.0.20.1	255.255.255.0	N/A
Router0	Gig0/0.100 (VLAN 100)	10.0.100.1	255.255.255.0	N/A
Router1	Gig0/1 (WAN)	10.0.12.2	255.255.255.252	N/A

Router1	Gig0/0.30 (VLAN 30)	10.0.30.1	255.255.255.0	N/A
Router1	Gig0/0.40 (VLAN 40)	10.0.40.1	255.255.255.0	N/A
Server0	Fa0 (VLAN 100)	10.0.100.10	255.255.255.0	10.0.100.1
PC0	Fa0 (VLAN 10)	DHCP	DHCP	10.0.10.1
PC1	Fa0 (VLAN 20)	DHCP	DHCP	10.0.20.1
PC2	Fa0 (VLAN 30)	DHCP	DHCP	10.0.30.1
PC3	Fa0 (VLAN 40)	DHCP	DHCP	10.0.40.1

VLAN Subnet Summary

VLAN ID	VLAN Name	Network Address	Gateway	DHCP Pool Name
10	Sales	10.0.10.0/24	10.0.10.1	VLAN10_POOL
20	IT	10.0.20.0/24	10.0.20.1	VLAN20_POOL
30	HR	10.0.30.0/24	10.0.30.1	VLAN30_POOL
40	Management	10.0.40.0/24	10.0.40.1	VLAN40_POOL
100	DNS_MGMT	10.0.100.0/24	10.0.100.1	VLAN100_POOL

COMMANDS

Switch0 Configuration

1. Create VLANs and assign access ports

```
Switch0(config)# vlan 10
Switch0(config-vlan)# name Sales
Switch0(config-vlan)# exit
Switch0(config)# vlan 20
```

```
Switch0(config-vlan)# name IT
Switch0(config-vlan)# exit
Switch0(config)# vlan 100
Switch0(config-vlan)# name DNS_MGMT
Switch0(config-vlan)# exit
```

2. Assign ports to VLANs

```
Switch0(config)# interface fastEthernet 0/2
Switch0(config-if)# switchport mode access
Switch0(config-if)# switchport access vlan 10
Switch0config-if)# exit
Switch0(config)# interface fastEthernet 0/3
Switch0(config-if)# switchport mode access
Switch0(config-if)# switchport access vlan 20
Switch0(config-if)# exit
Switch0(config)# interface fastEthernet 0/4
Switch0(config-if)# switchport mode access
Switch0(config-if)# switchport access vlan 100
Switch0(config-if)# exit
```

3. Configure trunk port to Router0

```
Switch0(config)# interface fastEthernet 0/1
Switch0(config-if)# switchport mode trunk
Switch0(config-if)# switchport trunk allowed vlan 10,20,100
Switch0(config-if)# exit
```

4. Verify

```
Switch0# show vlan brief
Switch0# show interfaces trunk
```

Switch1 Configuration

1. Create VLANs and assign access ports

```
Switch1(config)# vlan 30
Switch1(config-vlan)# name HR
Switch1(config-vlan)# exit
Switch1(config)# vlan 40
```

```
Switch1(config-vlan)# name Management
Switch1(config-vlan)# exit
```

2. Assign ports to VLANs

```
Switch1(config)# interface fastEthernet 0/1
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 30
Switch1(config-if)# exit
Switch1(config)# interface fastEthernet 0/2
Switch1(config-if)# switchport mode access
Switch1(config-if)# switchport access vlan 40
Switch1(config-if)# exit
```

3. Configure trunk port to Router1

```
Switch1(config)# interface fastEthernet 0/3
Switch1(config-if)# switchport mode trunk
Switch1(config-if)# switchport trunk allowed vlan 30,40
Switch1(config-if)# exit
```

Router0 Configuration

1. WAN interface

```
Router0(config)# interface gigabitEthernet 0/1
Router0(config-if)# ip address 10.0.12.1 255.255.255.252
Router0(config-if)# no shutdown
Router0(config-if)# exit
```

2. LAN sub-interfaces for VLANs

```
Router0(config)# interface gigabitEthernet 0/0
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# interface gigabitEthernet 0/0.10
Router0(config-subif)# encapsulation dot1Q 10
Router0(config-subif)# ip address 10.0.10.1 255.255.255.0
Router0(config-subif)# exit
Router0(config)# interface gigabitEthernet 0/0.20
Router0(config-subif)# encapsulation dot1Q 20
Router0(config-subif)# ip address 10.0.20.1 255.255.255.0
```

```
Router0(config-subif)# exit
Router0(config)# interface gigabitEthernet 0/0.100
Router0(config-subif)# encapsulation dot1Q 100
Router0(config-subif)# ip address 10.0.100.1 255.255.255.0
Router0(config-subif)# exit
```

3. DHCP pools with options

```
Router0(config)# ip dhcp excluded-address 10.0.10.1 10.0.10.10
Router0(config)# ip dhcp excluded-address 10.0.20.1 10.0.20.10
Router0(config)# ip dhcp pool VLAN10_POOL
Router0(dhcp-config)# network 10.0.10.0 255.255.255.0
Router0(dhcp-config)# default-router 10.0.10.1
Router0(dhcp-config)# dns-server 10.0.100.10
Router0(dhcp-config)# domain-name sales.local
Router0(dhcp-config)# lease 7
Router0(dhcp-config)# exit
Router0(config)# ip dhcp pool VLAN20_POOL
Router0(dhcp-config)# network 10.0.20.0 255.255.255.0
Router0(dhcp-config)# default-router 10.0.20.1
Router0(dhcp-config)# dns-server 10.0.100.10
Router0(dhcp-config)# domain-name it.local
Router0(dhcp-config)# lease 7
Router0(dhcp-config)# exit
Router0(config)# ip dhcp pool VLAN100_POOL
Router0(dhcp-config)# network 10.0.100.0 255.255.255.0
Router0(dhcp-config)# default-router 10.0.100.1
Router0(dhcp-config)# dns-server 10.0.100.10
Router0(dhcp-config)# lease 7
Router0(dhcp-config)# exit
```

4. EIGRP configuration

```
Router0(config)# router eigrp 100
Router0(config-router)# network 10.0.10.0 0.0.0.255
Router0(config-router)# network 10.0.20.0 0.0.0.255
Router0(config-router)# network 10.0.100.0 0.0.0.255
Router0(config-router)# network 10.0.12.0 0.0.0.3
Router0(config-router)# no auto-summary
Router0(config-router)# exit
```

Router1 Configuration

1. WAN interface

```
Router1(config)# interface gigabitEthernet 0/1
Router1(config-if)# ip address 10.0.12.2 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
```

2. LAN sub-interfaces for VLANs

```
Router1(config)# interface gigabitEthernet 0/0
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# interface gigabitEthernet 0/0.30
Router1(config-subif)# encapsulation dot1Q 30
Router1(config-subif)# ip address 10.0.30.1 255.255.255.0
Router1(config-subif)# exit
Router1(config)# interface gigabitEthernet 0/0.40
Router1(config-subif)# encapsulation dot1Q 40
Router1(config-subif)# ip address 10.0.40.1 255.255.255.0
Router1(config-subif)# exit
```

3. DHCP pools with options

```
Router1(config)# ip dhcp excluded-address 10.0.30.1 10.0.30.10
Router1(config)# ip dhcp excluded-address 10.0.40.1 10.0.40.10
Router1(config)# ip dhcp pool VLAN30_POOL
Router1(dhcp-config)# network 10.0.30.0 255.255.255.0
Router1(dhcp-config)# default-router 10.0.30.1
Router1(dhcp-config)# dns-server 10.0.100.10
Router1(dhcp-config)# domain-name hr.local
Router1(dhcp-config)# lease 7
Router1(dhcp-config)# exit
Router1(config)# ip dhcp pool VLAN40_POOL
Router1(dhcp-config)# network 10.0.40.0 255.255.255.0
Router1(dhcp-config)# default-router 10.0.40.1
Router1(dhcp-config)# dns-server 10.0.100.10
Router1(dhcp-config)# domain-name mgmt.local
Router1(dhcp-config)# lease 7
Router1(dhcp-config)# exit
```

4. EIGRP configuration

```
Router1(config)# router eigrp 100
Router1(config-router)# network 10.0.30.0 0.0.0.255
Router1(config-router)# network 10.0.40.0 0.0.0.255
Router1(config-router)# network 10.0.12.0 0.0.0.3
Router1(config-router)# no auto-summary
Router1(config-router)# exit
```

DNS Server Configuration

Static IP assigned on Server0 via Desktop > IP Configuration:

- IP Address: 10.0.100.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 10.0.100.1
- DNS Server: 10.0.100.10

DNS Records added via Services > DNS (DNS ON):

Record Name	**Type**	**IP Address**
router0.local	A Record	10.0.12.1
router1.local	A Record	10.0.12.2
pc0.sales.local	A Record	10.0.10.11
pc1.it.local	A Record	10.0.20.11
pc2.hr.local	A Record	10.0.30.11
pc3.mgmt.local	A Record	10.0.40.11
www.corp.local	A Record	10.0.100.10

VERIFICATION

EIGRP and DHCP Verification

On Router0 CLI:

```
show ip eigrp neighbors
show ip route
show ip dhcp binding
```

```
Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface      Hold Uptime    SRTT   RTO    Q   Seq
                                   (sec)          (ms)          Cnt  Num
0   10.0.12.2       Gig0/1         10   00:03:59  40     1000   0   7

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
C       10.0.10.0/24 is directly connected, GigabitEthernet0/0.10
L       10.0.10.1/32 is directly connected, GigabitEthernet0/0.10
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.1/32 is directly connected, GigabitEthernet0/1
C       10.0.20.0/24 is directly connected, GigabitEthernet0/0.20
L       10.0.20.1/32 is directly connected, GigabitEthernet0/0.20
D       10.0.30.0/24 [90/28416] via 10.0.12.2, 00:03:59, GigabitEthernet0/1
D       10.0.40.0/24 [90/28416] via 10.0.12.2, 00:03:59, GigabitEthernet0/1
C       10.0.100.0/24 is directly connected, GigabitEthernet0/0.100
L       10.0.100.1/32 is directly connected, GigabitEthernet0/0.100

Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.0.10.11       00E0.B01A.B60C           --                     Automatic
10.0.20.11       000A.4109.E180           --                     Automatic
Router#
```

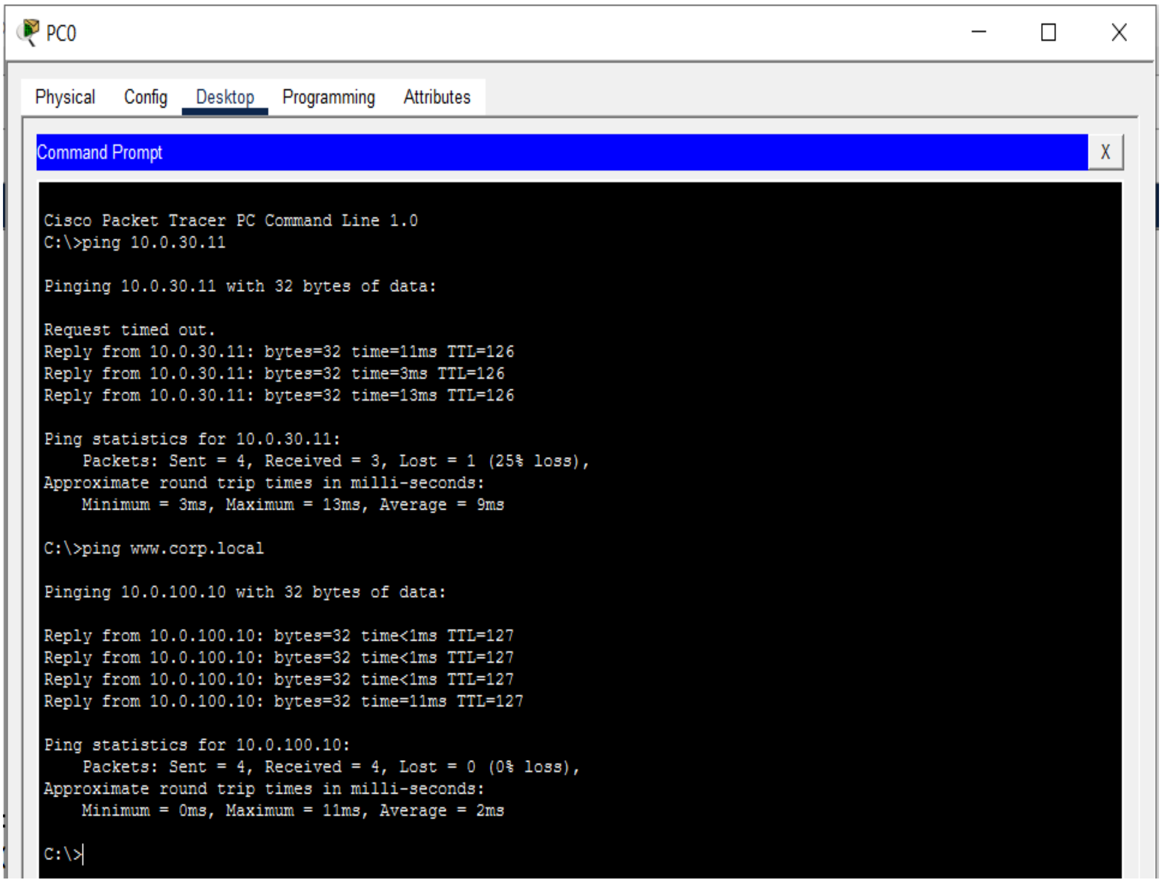
On Router1 CLI:

```
show ip eigrp neighbors
show ip route
```

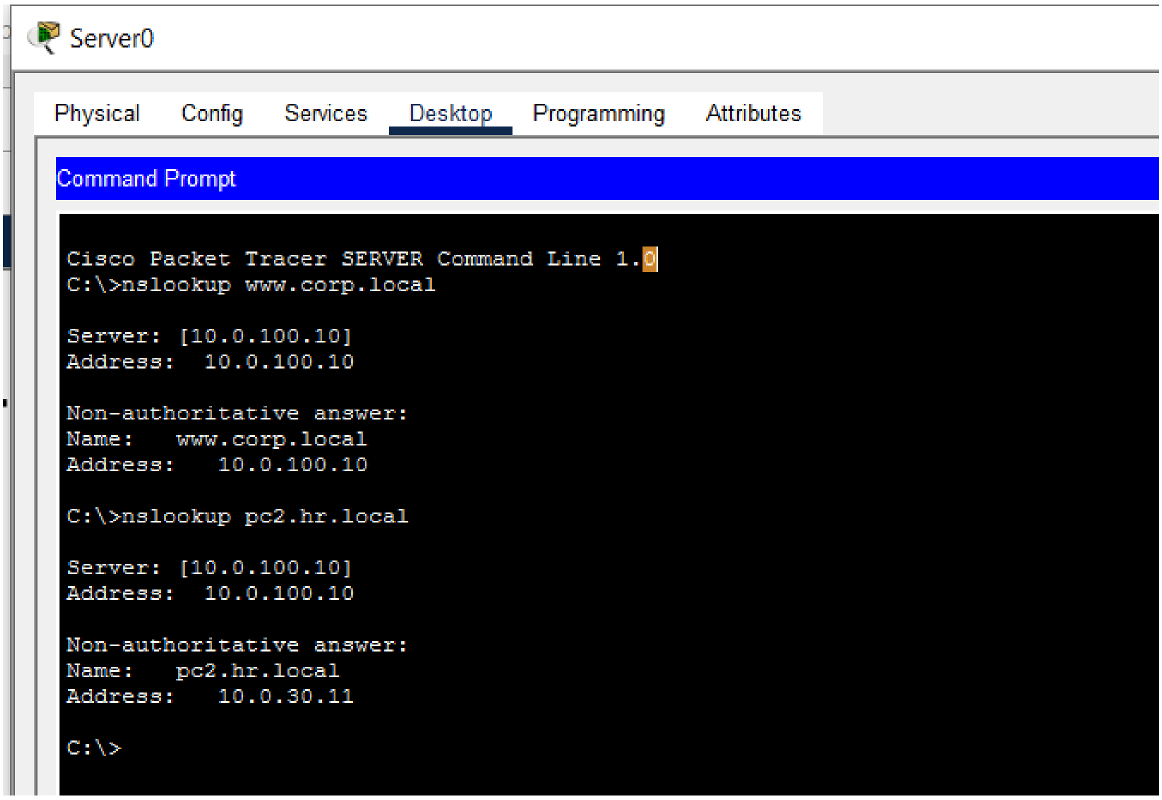
```
Router#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address         Interface       Hold Uptime    SRTT   RTO    Q    Seq
                                    (sec)          (ms)          Cnt  Num
0   10.0.12.1       Gig0/1          11   00:04:36  40     1000   0    5

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 9 subnets, 3 masks
D       10.0.10.0/24 [90/28416] via 10.0.12.1, 00:04:38, GigabitEthernet0/1
C       10.0.12.0/30 is directly connected, GigabitEthernet0/1
L       10.0.12.2/32 is directly connected, GigabitEthernet0/1
D       10.0.20.0/24 [90/28416] via 10.0.12.1, 00:04:38, GigabitEthernet0/1
C       10.0.30.0/24 is directly connected, GigabitEthernet0/0.30
L       10.0.30.1/32 is directly connected, GigabitEthernet0/0.30
C       10.0.40.0/24 is directly connected, GigabitEthernet0/0.40
L       10.0.40.1/32 is directly connected, GigabitEthernet0/0.40
D       10.0.100.0/24 [90/28416] via 10.0.12.1, 00:04:38, GigabitEthernet0/1

Router#
```

Ping Tests

From PC0 Command Prompt:



DNS Resolution (nslookup Simulation)



RESULTS

The project successfully implemented a multi-site network with two routers running EIGRP in Autonomous System 100. VLANs 10, 20, 30, and 40 were created and assigned to switch ports, along with VLAN 100 for DNS server management. IEEE 802.1Q trunking was configured between switches and routers, and inter-VLAN routing was achieved using router-on-a-stick with appropriate subinterfaces on both routers.

DHCP was configured on each router with option parameters including DNS server address, domain names, and lease duration. All PCs successfully received IP addresses from their respective VLAN pools.

EIGRP neighbor adjacency was established over the WAN link, and routing information was exchanged correctly between routers. The DNS server was configured with multiple records, and hostname resolution was verified using ping and nslookup. End-to-end connectivity across all VLANs was successfully achieved with no packet loss after initial ARP resolution.

DISCUSSION

This lab demonstrated the integration of multiple fundamental networking concepts within a single Cisco Packet Tracer topology. EIGRP was chosen as the dynamic routing protocol due to its fast convergence, use of DUAL algorithm, and support for VLSM, making it suitable for the multi-subnet design. The no auto-summary command was essential to prevent EIGRP from summarizing classful networks, which would have caused routing issues with the 10.0.0.0 address space shared across all subnets.

VLAN segmentation using 802.1Q trunking provided logical separation of departments while sharing physical infrastructure. The router-on-a-stick approach using subinterfaces allowed a single physical Gig0/0 port to route traffic between all VLANs efficiently. Placing the DNS server in a dedicated VLAN 100 with its own subinterface and DHCP pool ensured that management traffic was logically isolated from end-user traffic.

The DHCP option configuration, specifically the dns-server and domain-name options within each pool, ensured that clients not only received IP addresses but were also pre-configured to use the central DNS server and were assigned appropriate domain names based on their VLAN. This reflects real-world enterprise DHCP practices where option 6 (DNS server) and option 15 (domain name) are standard inclusions.

CONCLUSION

This project demonstrated the configuration of a network incorporating EIGRP dynamic routing, VLAN segmentation, 802.1Q trunking, DHCP with option parameters, and DNS with multiple records. The topology showed how routers, switches, and servers work together to provide a fully functional, segmented enterprise-like network. EIGRP enabled automatic route exchange between the two sites, DHCP simplified host configuration, and DNS provided name resolution across all VLANs. Verification through show commands, ping tests, and nslookup simulation confirmed correct operation of all configured services.